

An Efficient CatBoost-based Model for Hospital Length of Stay Prediction

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Abstract— A proper ICU LOS is one of the main inpatient care and hospital resource management. One of the traditional machine learning-based LOS prediction schemes is supplemented with the CatBoost method that is effective in utilizing categorical EHR data and exhibits no overfitting. Hospital stays are used to train a longer model, which is then evaluated for accuracy, precision, recall, F1-score, and AUC. CatBoost has a higher precision of predictions when compared to the common classifiers. To enhance the transparency of models, explainable artificial intelligence (XAI) algorithm like SHAP are applied to investigate and compare clinical variables affecting the length of ICU stay. The flask deployment service offers web-based and real-time ICU stays prediction. The suggested expansion is used to clarify why CatBoost and explainable machine learning can predict the duration of stay in intensive care unit in the modern healthcare in a way that is accurate, understandable and effective.

Keywords— Hospital bed management, ICU length of stay, electronic health records (EHR), machine learning, XGBoost, CatBoost, SHAP, explainable artificial intelligence (XAI), prediction models, healthcare analytics.

I. INTRODUCTION

The use of prediction of LOS in hospitals and in intensive care units maximizes the utilization of resources, lowers costs of activities and enhances patient outcomes. Recent studies have shown that ML models are more accurate than statistical techniques that use EHR in predicting hospital length of stay (LOS). The explainable machine learning plan of lung cancer LOS prediction by the efforts of Alsinglawi et al. [1] dwells on the interpretability of clinical decisions. As Alsinglawi [2] observed, smart predictive analytics can be used to optimize the consumption of hospital resources through EHR data.

Clinical and diagnostic data have been used to predict LOS in several researches. Staziaki et al. [3] Trauma patients' hospital stays and intensive care unit admissions were

predicted by machine learning that added clinical characteristics to CT images. Su et al. used machine learning models to assess the severity, mortality, and length of stay of ICU sepsis patients. [4] showed that ensemble learning in critical care works. These approaches have issues with category norms and are likely to make use of numerical clinical indicators.

Alghatani et al. [5] using machine learning methods to forecast LOS based on vital sign data; the algorithms performed rather well, but they had low interpretability and feature transparency. Gradient Boosting and Random Forest are both effective predictors, however, it can be challenging to keep a variety of EHR data and provide physicians with projections that are easy to understand. This study enhances LOS prediction frameworks by incorporating the CatBoost algorithm, which successfully handles categorical data and eliminates overfitting, and Explainable AI techniques to increase interpretability and clinical relevance.

II. LITERATURE SURVEY

Based on a previous study, increased bed occupancy levels in an inpatient hospital that results in early hospital discharges that prioritize more on the need of hospitals admitting more ED patients at the expense of the medical needs of the patients. We reckon that the unintended outcome would be more unplanned readmissions with patients allowed out of hospital since the hospital was crowded. As a result, 30-day readmission rates after hospital discharge are compared to inpatient beds.

Methods: All 2011–2012 emergency department admissions from a 420-bed southern Swedish hospital disclosed before December 1, 2012, are included. After being released, the unexpected readmission rate (30 days) was determined at less than 95, 95–100, 100–105, and more than

105 inpatient bed occupancy rates. The binary logistic regression was built to take into account age, time of discharge among other variables.

Findings: the results were acquired based on 32,811 total visits out of which 9.9% resulted in unexpected readmission in 30 days of discharge. Discharge occupancy of 95 or less was 9.0% readmission, 10.2% was 95-100, 10.8% was 100-

105, and 10.5% was over 105. For patients released at 95- 100 occupancy, the OR (95% CI) of readmission was 1.11 (1.01-1.22), 1.17 (1.06-1.29) for 100-105 occupancy, and

1.15 (0.99-1.34) for greater occupancy than 105, according to the multivariate models.

Conclusions: The unplanned 30-day readmission rates were higher in the said patients who were discharged at the inpatient wards when the bed occupancy was high [6].

The number of patients and budget constraints pose a challenge to clinical workers in coping with the hospital bed. The most significant among them is efficient distribution of ICU beds that have a heavy workload to the life-support patients. To address this issue, find out how long existing ICU patients have been there. Using temporal and pointwise (1x1) convolution on the eICU and MIMIC-IV datasets, To forecast the length of critical care, we present a new deep learning model. Missing data, unequal temporal sampling, and EHR skewness are all fixed via Temporal Pointwise Convolution (TPC). Our performance was 18–68% better than that of the well-known LSTM network and the Transformer multi-head self-attention network (metric and dataset dependent). Mortality prediction is included as a side task to improve performance. The remaining stay's mean absolute variation is 2.28 days (MIMIC-IV) and 1.55 days (eICU) [7].

Purpose: To determine whether providing essential services in a structured, supervised manner on-site enhances care while using less.

Design: Over time, two surgical intensive care units were examined. The research population was treated by an on-site critical care team under the supervision of an intensivist. Under the direction of a general surgeon, multi-site staff oversaw the control group. The longest hospital stay, resource consumption, and complications mattered most.

Setting: The patients were academic Medical Centre and general surgery patients.

Findings: Because they spent less time in the surgical critical care unit, used fewer resources, experienced fewer problems, and spent less money in the hospital, critical care patients had high Acute Physiology and Chronic Health Evaluation II ratings. APACHE II patients with the lowest scores changed the most.

Conclusions: Treatments for critical care are costly and dangerous. Techniques that are coordinated and verified are needed to analyze the supply of this resource. The critical care service model did well in this study in terms of both cost and quality [8].

APACHE II (acute physiology and chronic health assessment) is a tool that is extensively applicable to predicting the hospital mortality and ICU stay. APACHE II forms and scores were given to every resident of ICU within 24 hours of admission. A low score below 10 on APACHE II

score represents a short ICU stay and 100 percent likelihood of being discharged (by international standards). Within the 24 hours of admission, the uniform scoring system has the ability to predict mortality, length of stay, resource use, antibiotic use, and ethical concerns of end-of-life counseling of families [9].

The untrustworthy and efficient classification approach limits the use form and intensive care worth research. Acute physiology and chronic health assessment (APACHE) for critically sick patients was validated in this study. APACHE utilizes medical record information. APACHE was able to categorise the ICU admissions of 223 community hospital ICU admissions, and 582 university hospital ICU admissions. Responses to the APACHE acute physiology score were equal in validation experiments of these 805 admissions to the treatment effort and mortality. This was true in the case of the various groups of patients utilizing sensitivity analysis. As soon as the APACHE classification system is multi-institutional validated, it will be able to make adjustments on the basis of the case mix, compare outcomes, review new drugs, and measure the use of ICU [10].

III. METHODOLOGY

The boosting algorithm and the explainable machine learning are applied to predict the length of stay in the ICU of the extended approach. Data visualization reveals patterns and distribution of features in Electronic Health Records (EHR) of patients. Data preparation is used to stabilize models by encoding categorical variables or filling in missing values, and standardizing numerical attributes. Multiple base models are trained in order to compare them in training and evaluating sets of the processed dataset. CatBoost is an ordered boosting algorithm which works with categorical data and reduces overfitting. Model performance evaluation is done in terms of F1-score, accuracy, precision, and recall. Clinical reliability and transparency of prediction are ensured by explainable AI algorithms that decode the input of features.

Since CatBoost is an efficient model in handling categorical health data, which does not require any preprocessing, it was selected as the primary model. In comparison to gradient boosting, CatBoost applies target statistics and ordered boosting to reduce prediction bias and overfitting.

Some of the categorical elements provided by the EHRs include hospital type, ward details, and admission category. The CatBoost auto encodes the categorical data during the training and therefore enhancing the precision and stability of the model. CatBoost uses clinical data to accurately forecast the length of an ICU stay.

In order to prevent data leakage in the model training, the data was first segmented into training and testing to adhere to model evaluation processes. The training data were directly applied in model learning, and the testing data were not visible during model learning.

All the preprocessing procedures like normalization and the feature encoding were done in a uniform manner to both subsets but statistical parameters were only obtained using the training data. This will avoid using information of the test set to inform the training process, which will mean the evaluation of the models remains unbiased.

A) Proposed System

The proposed study utilizes the newest machine learning and explainable AI algorithms to improve the clinical understandability and ICU LOS's capacity for prediction. The system profiles ICU stay as being short or lengthy at admission with the use of patient EHR records, including demographics, clinical conditions, and admission related data. In order to carry out a comparative analysis, various simple machine learning models are employed, but the biggest multiplication denotes that the CatBoost model should be utilized due to the possibility to operate with categorical healthcare figures and reduce overfitting. It enables the system to provide the proper LOS estimations of the bed management and hospital resource planning.

Explainable Artificial Intelligence (XAI) will be used in the proposed study to visualize model alternatives and increase forecast accuracy. FCA could help the clinicians to determine the factors of the ICU stay duration, contributing to the greater credibility and usefulness of real-world healthcare. The system takes the benefit of Flask, which offers a plain web interface to forecast the weather in real-time using LOS. This is a hospital-wide, end-to-end solution, which enhances patient care and productivity within the hospital because the length of stay is predicted conveniently, interpretively, and accurately in the intensive care unit.

B) System Architecture

The flexible and scalable pipeline uses explainable machine learning to forecast ICU stays. First patient EHRs are found in one repository because they are in the hospital stay data. Exceptionally a preprocessing layer is used to make data more presentable, attribute names are categorised, missing values are processed and numbers are normalized. Testing and training data are kept apart in order to model assessment after impartial preprocessing. Unidirectional data flow enables a speedy learning process and equal performance evaluation of machine learning models.

The architecture has numerous classifiers in the model training and prediction layer and the CatBoost model with extended version is highlighted. CatBoost works well with category data and this can boost prediction accuracy and minimize overfitting. The accuracy, precision, recall, and F1-score of trained models for ICU stay prediction are assessed. The XAI process model output's explainability layers and identify which factors can be contributing significantly. Finally, the system web interface is implemented using

Flask, which allows clinicians to forecast and view the outcome in real-time so that they can make decisions and control resources.

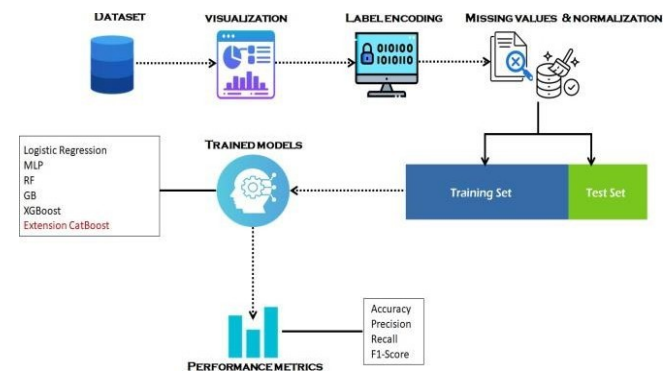


Fig.1. Proposed architecture

IV. IMPLEMENTATION

A) MODULES

a. Data Collection Module

The module will collect patient Electronic Health Records (EHR) based on hospital stay data set. It gathers demographic, clinical and admission related attributes that are required in prediction as far as ICU length of stay is concerned.

b. Data Visualization Module

The module plots and analyses the dataset, so as to learn the distribution, correlation and pattern of features. Visualization will help to identify the outliers and data imbalance before the models are trained.

c. Data Preprocessing Module

This package standardizes the numeric features, codifies the categorical ones by label encoding as well as the missing data. Preprocessing enhances the consistency of the data and model performance.

There were a number of preprocessing steps done before model training in order to guarantee data quality. Numerical variable standardization, nominal variable labeling, and missing values are all involved.

Label encoding transformed category variables into numerical values to aid in the interpretation of data by machine learning models. To normalize numerical attributes in order to enhance training stability, feature scaling was used.

d. Dataset Splitting Module

Training and testing data were obtained through processing. In order to evaluate machine learning models subjectively, this module separates learning and validation data.

e. Model Training Module

Data is trained using Logistic Regression, Random Forest, MLP, Gradient Boosting, XGBoost, and extended CatBoost. One disadvantage of CatBoost is its efficacy with categorical data.

f. Prediction Module

This module gives the forecasts of ICU stay depending on the short and long stay groups of patients. The trained models are utilized to implement on unknown test data to find out the accuracy of the prediction.

g. Explainability Module (XAI)

To change model results, the section uses Explainable AI, which is like SHAP. By identifying ICU stay factors, it increases provider confidence and openness.

h. Performance Evaluation Module

F1-score, recall, accuracy, and precision are used to gauge the model's performance. Baseline and improved CatBoost models are compared in this module.

i. Deployment Module (Flask Web Interface)

The last module presents the learnt model with the use of Flask. ICU duration of stay and result prediction are made simple by its web interface.

B) ALGORITHMS

a. Logistic Regression

As a baseline of predicting ICU duration of stay, the method of predicting ICU duration of stay is Logistic Regression, which is a supervised learning method. It estimates the likelihood of a patient to spend the short or the long stay at the ICU based on the input EHRs. The strategy is computationally active and easy to comprehend, so that is why it is reasonable to compare it at the initial stage. However, a very low score on complicated medical data is as a result of its linear decision boundary. However, it provides a great basis upon which complicated models may be evaluated.

b. Random Forest

Numerous decision trees are used in Random Forest ensemble learning predictions. It is helpful in identifying non-linear correlations between ICU stay and patient variables. The strategy reduces overfitting and is also resistant to noise through bootstrap aggregation. On the level of individual choices, however, it can be difficult to be interpretable. It can be considered an effective traditional group base of our study.

Random Forest inhibits overfitting in bootstrap aggregation (bagging). In this process, the decision trees are trained on random sampled data.

One thing is predicted by each tree and the overall result is acquired due to averaging predictions of all the trees. Such an approach to assembling assists in reducing the variance and prevents the model to be overfitted to some tendencies of training data.

c. Multi-Layer Perceptron (MLP)

Multi-layer perceptrons refer to artificial neural networks that are feedforward, i.e. they have input, hidden and output layers. It is capable of training on complicated and high-dimensional trends on patient EHR data. MLP is predictive and it learns the interaction of features which are not simple. But it lacks internal explainability and needs to be hyperparameter optimized. That is why neural models have been applied with XAI schemes.

d. Gradient Boosting

Gradient Boosting creates the models in a process that amends the shortcomings of the earlier models. It gets a high prediction level as it prioritizes on patient information which is hard to categorize. The algorithm can also deal with various types of features and interactions that are complicated. However, it may be overfitted when overfitted. Gradient Boosting provides a competitive level of ICU LOS predictor.

e. XGBoost

XGBoost is an improved and scalable version of gradient boosting that improves the result by balancing and employing parallel computation. It has the ability to work on large volumes of EHR within a short amount of time and it is more precise compared to other conventional methods of boosting. Besides, XGBoost is partially interpretable since it can perform an analysis of feature significance. It, however, has extremely specific parameter adjustments. XGBoost on the list of the sophisticated models in this study is a decent placebo.

f. CatBoost (Extension)

CatBoost is an advanced gradient boosting method that works with category data without the need for preprocessing. Ordered boosting enhances complex healthcare data generalization by lowering overfitting. CatBoost offers the greatest extended system prediction accuracy. Furthermore, it seamlessly integrates with Explainable AI methodologies to elucidate feature contributions. CatBoost is the core algorithm used in the proposed ICU duration of stay prediction extension.

The reason why the catBoost is selected as the key model is that it can successfully handle categorical healthcare data without necessarily carrying out significant preprocessing. Unlike classical methods of gradient boosting, CatBoost currently employs both ordered boosting and target statistics to reduce bias in predictions and overfitting significantly.

EHR are also healthcare data sets which contain several categorical variables e.g., type of hospital, ward, and the nature of admission. To improve the model quality and stability, CatBoost will automatically encode these categorical variables automatically. This makes CatBoost very suitable in the prediction of the ICU length of stay problems with heterogeneous clinical data.

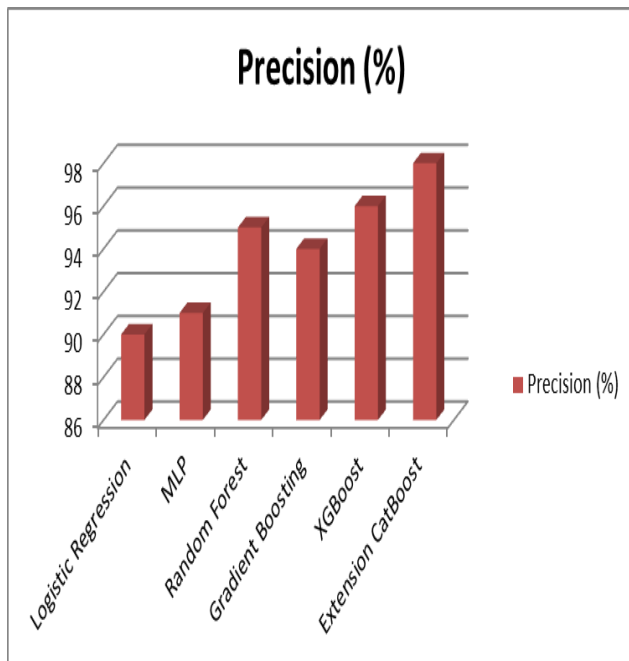
C) About Dataset

The experiments did were with Kaggle Hospital Length of Stay. The data will consist of 1,00,000 EHR hospital admissions. The records contain the hospital and medical information on the admission of a patient respectively.

The medical operations, health concerns, hospitalization and demographics determine the number of days spent at the ICU. Some of the input factors in the dataset are age group, hospital code, type of admission, severity of the illnesses, department and ward.

The last column is the variable of interest which shows short or long ICU stays. These labels are used to forecast ICU stay duration and train and assess machine learning algorithms.

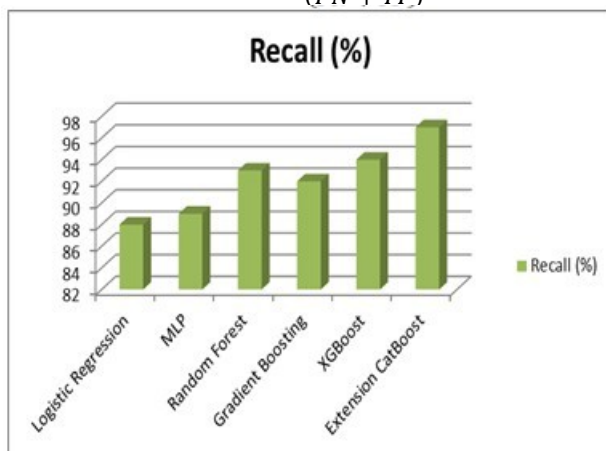
The author trained and tested the algorithms using the real-time AIN hospital dataset, however as it is not publicly



Recall: The ability of a model to identify all pertinent machine learning classes is known as recall. The capacity of a model to recognize class instances is demonstrated by the ratio of correctly predicted positive observations to total positives.

$$\text{Recall} = \frac{TP}{(TP + FN)}$$

$$\text{Recall} = \frac{TP}{(FN + TP)}$$



F1-Score: ML model accuracy is shown by a high F1 score. Model accuracy is increased when recall and accuracy are combined. The accuracy statistic displays how often a model predicts a dataset properly.

$$F1 \text{ SCORE} = 2 / ((1/PRECISION) + (1/RECALL))$$

$$F1 \text{ SCORE} = (2 * PRECISION * RECALL) / (PRECISION + RECALL)$$

$$F1 = 2 \cdot \frac{(\text{Recall} \cdot \text{Precision})}{(\text{Recall} + \text{Precision})}$$

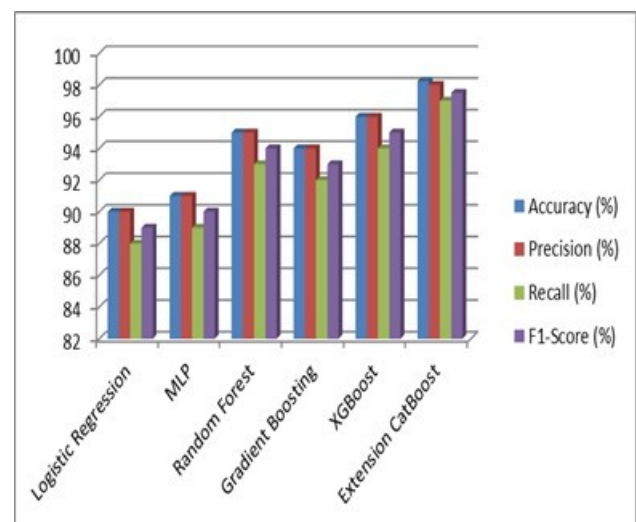
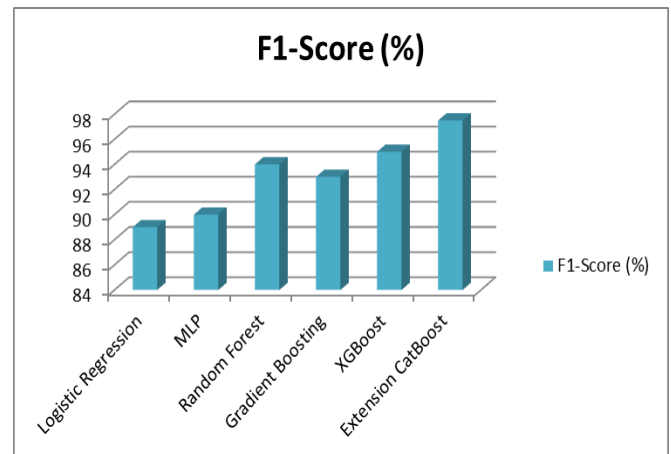


Fig 2 Accuracy over Epochs

Machine learning models are compared using accuracy, precision, recall, and F1-score in Fig. 2. The logistic Regression and the MLP use cases show moderate predictive ability because of the weak capability to represent the complex relationships in EHR data.

Ensemble models such as the RF and the GB have better performance because they use a combination of many decision trees. XGBoost is the most accurate of the baseline models because it uses regularization and efficient implementation of gradient boosting. The overall performance of the extended CatBoost model is best, which proves to be effective with respect to the use of categorical features and underfitting.

Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Logistic Regression	90	90	88	89
MLP	91	91	89	90
Random Forest	95	95	93	94
Gradient Boosting	94	94	92	93
XGBoost	96	96	94	95
Extension CatBoost	98.25	98	97	97.5

Table 1 ML Models performance

VI. CONCLUSION

According to this article, the extended ICU length of stay prediction model based on CatBoost is more accurate and cognizant than the current machine learning models. CatBoost was more superior to all the baseline algorithms based on key performance metrics in that it controlled categorical EHR data and minimized overfitting. The feature contributions, as shown by explainable AI, boosted the trust of the clinicians and decision-making. The practical model implementation and use in hospitals and real-time is provided on a web interface that is flask based. The suggested expansion will offer useful, trustworthy, and understandable ICU resources management and patient care optimization.

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